Gradient Boosting and LSTM/RNN Model Comparison

This report compares the two predictive models using the verified held-out test period from 2026-04-16 00:00:00 to 2026-05-01 00:00:00. These metrics are based on real measured values, so they are used for model comparison. The June outputs are forecasts only and are not scored here.

Metric Results

Model	Target	MAE	MSE	RMSE	R2	Mean Relative Error
Gradient Boosting	Temperature	0.3681	0.2656	0.5153	0.9873	2.46%
Gradient Boosting	PM10	1.5291	5.2856	2.2990	0.7758	10.01%
LSTM/RNN	Temperature	0.4977	0.4120	0.6419	0.9803	3.57%
LSTM/RNN	PM10	1.6206	5.3221	2.3070	0.7742	10.97%

Error Margin Difference

Target	Metric	Gradient Boosting	LSTM/RNN	Difference
Temperature	MAE	0.3681	0.4977	0.1296
Temperature	RMSE	0.5153	0.6419	0.1265
Temperature	Mean Relative Error	2.46%	3.57%	1.11%
PM10	MAE	1.5291	1.6206	0.0916
PM10	RMSE	2.2990	2.3070	0.0079
PM10	Mean Relative Error	10.01%	10.97%	0.96%

Interpretation

For temperature, Gradient Boosting had MAE 0.3681 C and RMSE 0.5153 C, while LSTM/RNN had MAE 0.4977 C and RMSE 0.6419 C. The lower MAE and RMSE show that Gradient Boosting made smaller temperature errors on the verified test set.

For PM10, Gradient Boosting had MAE 1.5291 ug/m3 and RMSE 2.2990 ug/m3, while LSTM/RNN had MAE 1.6206 ug/m3 and RMSE 2.3070 ug/m3. The PM10 difference is small, which means both models captured a similar amount of the PM10 pattern in the test period.

R2 is closer to 1.0 for better fit, while MAE, MSE, RMSE, and mean relative error are better when lower. Overall, Gradient Boosting performed slightly better on the verified data, especially for temperature. LSTM/RNN remained competitive for PM10, but its recurrent structure did not give a clear advantage on this dataset.

Metric Definitions

MAE is the average absolute size of the prediction error. MSE is the average squared error and penalizes larger mistakes more strongly. RMSE is the square root of MSE, so it is easier to read in the original unit of the target. R2 measures how much variation is explained by the model; values closer to 1 are better.